# Secure Network Architecture Report

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# Network Segmentation Goals and Access Controls

## Purpose of this Network Configuration

The purpose of this network configuration was to implement secure network segmentation between the Finance, HR, and Database subnets. The objective was to allow authorized Finance systems to access the MySQL service hosted on the Database Server while preventing unauthorized access from the HR subnet.

Network segmentation is necessary because it reduces the attack surface, limits lateral movement by attackers, protects sensitive information, and enforces the principle of least privilege. By separating systems into distinct security zones, organizations can better control communication between resources and reduce the impact of a security breach.

### Access Controls Implemented

The following access controls were implemented:

- Dedicated Database subnet (192.168.3.0/24)
- DHCP configuration for dynamic IP assignment
- Firewall rules restricting MySQL access
- Access based on subnet membership
- Traffic monitoring using tcpdump
- Deny-by-default security model

### Alignment with Security Best Practices

This configuration aligns with several security frameworks:

- PCI DSS Requirement 1 (Network Segmentation and Firewall Controls)
- CIS Controls v8 (Network Infrastructure Management)
- Principle of Least Privilege
- Defense-in-Depth Security Strategy
- Zero Trust Network Concepts

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## Access Control Rules and Justification

### Allowed Traffic

- Finance subnet (192.168.1.0/24) allowed access to Database subnet (192.168.3.0/24) on TCP port 3306 (MySQL).

### Denied Traffic

- HR subnet (192.168.2.0/24) denied access to Database subnet on TCP port 3306.
- All unauthorized MySQL traffic blocked.

### Security Justification

These rules ensure only users and systems with a legitimate business need can access financial data. Restricting database access minimizes the risk of unauthorized access, data breaches, and lateral movement within the network.

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# Firewall Rules and Network Traffic Control

## Firewall Configuration and Implementation

The firewall was configured on the VyOS router to enforce communication restrictions between network segments.

## Network Segments

| Segment | Subnet |
|---|---|
| Finance | 192.168.1.0/24 |
| HR | 192.168.2.0/24 |
| Database | 192.168.3.0/24 |

## Implemented Firewall Rules

| Rule Name | Source Subnet | Destination Subnet | Protocol | P | Action |
|---|---|---|---|---|---|
| FIN-DB-ALLOW | 192.168.1.0/24 | 192.168.3.0/24 | TCP | 3 | Ac<br>c<br>e |

| | | | | | | |
|---|---|---|---|---|---|---|
| | | | | | | p<br>t |
| HR-DB-DEN<br>Y | 192.168.2.<br>0/24 | 192.168.3.0/24 | TCP | 3 | Dr | c<br>p |
| DEFAULT-D<br>ENY | Any | 192.168.3.0/24 | Any | A | Dr | c<br>p |

### Rule Justification

**FIN-DB-ALLOW**
Allows Finance users to access the MySQL database required for financial operations.

**HR-DB-DENY**
Prevents HR systems from accessing sensitive financial databases.

**DEFAULT-DENY**
Blocks all unspecified traffic to maintain a secure posture and prevent unauthorized access.

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# Traffic Validation and Analysis

## Network Connectivity Tests

Connectivity testing was performed using ping and MySQL connection attempts.

### Finance VM Results

- Successfully reached the Database Server.
- Successfully connected to MySQL service on port 3306.

- Firewall allowed traffic as expected.

### HR VM Results

- Could not establish a MySQL connection.
- Firewall correctly blocked access.
- Segmentation policies functioned properly.

### Troubleshooting Performed

The following checks were completed:

- Verified IP addresses using `ip a`
- Verified routing tables using `ip route show`
- Confirmed VyOS forwarding configuration
- Verified MySQL service status
- Confirmed firewall rule order and priority

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## Capturing and Analyzing Traffic

### Role of tcpdump

Tcpdump was used to capture and analyze network traffic between subnets. It provided visibility into allowed and denied connections and verified firewall enforcement.

Example command:

- sudo tcpdump -i ens3 port 3306

### Observed Traffic Patterns

#### Expected Behavior

- Finance subnet traffic successfully reached the database.
- MySQL sessions established correctly.
- TCP handshakes completed.

#### Denied Traffic

- HR subnet traffic attempted connection.
- TCP SYN packets observed.
- No successful connection established.
- Firewall dropped traffic.

### Example Captured Data

| Source IP | Destination IP | Protocol | P | Allowed/Blocked |
|---|---|---|---|---|
| 192.168.1.2 | 192.168.3.2 | TCP | 3 | Allowed |
| 192.168.2.2 | 192.168.3.2 | TCP | 3 | Blocked |

### Unexpected Traffic

No unexpected traffic was detected during testing. All observed traffic matched the intended security policy.

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# Reflection on Network Security Implementation

## Effectiveness of Network Segmentation

The segmentation rules successfully enforced isolation between departments and protected the database subnet. Only authorized systems were able to access sensitive services.

Firewall rules provided an effective mechanism for enforcing access control and limiting communication paths between network segments.

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## Challenges and Adjustments

Several challenges were encountered during implementation:

- Missing routes between subnets.
- Incorrect firewall rule ordering.
- Initial MySQL connection failures caused by service configuration.

These issues were resolved by:

- Verifying routing tables.
- Reordering firewall rules.
- Restarting network and database services.
- Testing connectivity after each configuration change.

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## Future Security Enhancements

Additional security improvements could include:

- Multi-Factor Authentication (MFA)
- Network Access Control (NAC)
- Intrusion Detection and Prevention Systems (IDS/IPS)
- TLS encryption for MySQL traffic
- Security Information and Event Management (SIEM)
- Zero Trust architecture

### Logging and Monitoring Improvements

- Centralized log collection
- Automated alerting
- Firewall logging

- Continuous traffic monitoring
- Security event correlation through SIEM

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# Broader Security Implications

Network segmentation is critical in enterprise environments because it reduces the attack surface and limits the impact of compromised systems.

Controlled access points improve cybersecurity resilience by ensuring that only authorized systems can communicate with sensitive resources.

Segmentation can help mitigate attacks such as:

- Ransomware lateral movement
- Insider threats
- Credential compromise attacks
- Database breaches
- Malware propagation
- Unauthorized network reconnaissance

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## Conclusion

This lab successfully demonstrated how secure network segmentation and firewall controls can protect sensitive systems while allowing necessary business communication. The Finance subnet was granted controlled access to the Database Server, while HR access was effectively blocked. Traffic validation and packet analysis confirmed that the implemented security controls functioned as intended and aligned with industry security best practices.

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